

AI-Powered Supply Chain Knowledge and Decision Support System

Overall Project Objective

To develop an AI-powered platform that helps supply chain professionals, managers, and students access organizational knowledge quickly and make better operational decisions using artificial intelligence, retrieval systems, analytics, and decision support algorithms.

PHASE 1: Intelligent Supply Chain Knowledge Assistant with Dashboard

Phase Objective

Build an intelligent knowledge retrieval system that understands supply chain questions, searches trusted organizational documents, and returns accurate answers with citations and supporting evidence.

Phase 1 Deliverables

1. Supply Chain Knowledge Repository

A centralized document repository containing:

- Standard Operating Procedures (SOPs)
 - Procurement policies
 - Inventory management policies
 - Warehouse operation manuals
 - Transportation guidelines
 - Vendor contracts
 - Shipping procedures
 - Customs and import/export regulations
 - ERP user manuals
 - Supply chain textbooks
 - Research papers
 - Government policy documents
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2. Document Processing Pipeline

The system should be able to:

- Upload PDF documents
- Extract text from documents
- Split documents into smaller chunks
- Clean and preprocess text
- Store processed information for retrieval
- Maintain document metadata

Metadata includes:

- Document name
 - Section title
 - Page number
 - Document category
 - Upload date
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3. Vector Database

A searchable knowledge database containing:

- Embedded document chunks
 - Metadata information
 - Similarity search functionality
 - Fast retrieval capability
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4. Retrieval-Augmented Generation (RAG) Engine

The system should:

- Accept natural language questions.
- Search relevant knowledge chunks.
- Retrieve the most relevant information.
- Generate answers only from retrieved content.

- Prevent hallucination or fabricated information.
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5. Citation-Based Answer Generation

Every answer must include:

- Source document name
- Page number
- Retrieved passages
- Confidence score
- References used to generate the answer

Example:

Question:

"What is the recommended reorder policy for raw materials?"

Answer:

"The inventory policy recommends maintaining 14 days of safety stock and using a reorder point system."

Source:

Inventory Policy Manual

Page:

27

Confidence:

92%

6. Intelligent Query Interface

Users can ask questions such as:

- What is the reorder policy?
 - Which suppliers are approved?
 - What is the lead time for imported materials?
 - What are the customs procedures for exports?
 - What is the warehouse receiving process?
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7. Role-Based User Interface

Possible user roles:

- Supply Chain Manager
 - Procurement Officer
 - Warehouse Supervisor
 - Logistics Coordinator
 - Student or Research User
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8. Interactive Dashboard

Dashboard should include:

Document Analytics

- Number of uploaded documents
- Documents by category
- Recent uploads
- Most accessed documents

Query Analytics

- Number of questions asked
- Frequently asked questions
- Most searched topics
- User activity statistics

Knowledge Coverage Analytics

- Coverage by supply chain function
 - Missing knowledge areas
 - Document completeness indicators
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9. Source Traceability Module

The system must show:

- Retrieved document snippets

- Highlighted source paragraphs
 - Citation locations
 - Retrieval ranking
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10. Performance Evaluation

Evaluation metrics:

- Retrieval Accuracy
 - Precision
 - Recall
 - Response Time
 - User Satisfaction Score
 - Citation Accuracy
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11. Technical Deliverables

- Complete source code
 - Database schema
 - API documentation
 - Deployment guide
 - Installation guide
 - User manual
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12. Academic Deliverables

- Literature review
- System architecture diagram
- Data flow diagram
- Methodology chapter
- Experimental results
- Phase 1 report

- Phase 1 presentation
 - Demonstration video
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PHASE 2: Intelligent Supply Chain Decision Support System

Phase Objective

Transform the knowledge assistant into an intelligent system that not only answers questions but also recommends actions and supports managerial decisions.

Phase 2 Deliverables

1. Decision Support Engine

The system should analyze:

- Inventory levels
- Demand information
- Lead times
- Supplier performance
- Transportation constraints
- Service levels

and provide recommendations.

2. Inventory Decision Module

Capabilities:

- Reorder point recommendation
- Safety stock calculation
- EOQ recommendation
- Inventory risk alerts
- Overstock detection
- Stockout prediction

Example output:

"Current inventory is expected to run out in 9 days. Recommended order quantity is 1,250 units."

3. Supplier Decision Module

Capabilities:

- Supplier comparison
 - Supplier scorecards
 - Risk evaluation
 - Vendor ranking
 - Lead time analysis
 - Delivery performance tracking
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4. Procurement Decision Module

Capabilities:

- Purchase timing recommendation
 - Supplier selection recommendation
 - Procurement risk identification
 - Alternative supplier suggestion
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5. Transportation Decision Module

Capabilities:

- Transport mode recommendation
 - Route comparison
 - Cost analysis
 - Delivery time estimation
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6. Demand and Forecasting Module

Capabilities:

- Demand forecasting
 - Trend analysis
 - Seasonal demand identification
 - Forecast accuracy measurement
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7. Risk Monitoring Module

Capabilities:

- Supply disruption alerts
 - Supplier risk alerts
 - Demand variability alerts
 - Inventory shortage alerts
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8. Scenario Simulation Module

Users can simulate:

- Supplier delays
- Demand spikes
- Port disruptions
- Transportation failures
- Inventory shortages

Example:

"What happens if supplier lead time increases by 30%?"

The system estimates:

- Stockout probability
 - Additional inventory required
 - Service level impact
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9. Recommendation Engine

The system should recommend actions such as:

- Increase safety stock by 15%
 - Shift orders to Supplier B
 - Use air freight instead of sea freight
 - Advance purchase orders by 7 days
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10. Advanced Decision Dashboard

Dashboard components:

Inventory Health Dashboard

- Inventory turnover
- Days of inventory
- Service level
- Stockout risk

Supplier Dashboard

- On-time delivery
- Defect rates
- Lead time variability
- Supplier ranking

Procurement Dashboard

- Purchase trends
- Spend analysis
- Supplier concentration

Logistics Dashboard

- Transportation cost
 - Delivery performance
 - Route efficiency
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11. Explainable AI Module

Every recommendation should explain:

- Why the recommendation was generated
 - Which data was used
 - Expected benefits
 - Associated risks
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12. AI Agent Capability (Optional Advanced Feature)

The AI agent can:

- Answer questions
 - Analyze data
 - Generate recommendations
 - Compare alternatives
 - Explain decisions
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13. Integration Capability

Future integration possibilities:

- ERP systems
 - Warehouse Management Systems
 - Procurement systems
 - Inventory databases
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14. Academic Deliverables

- Decision support methodology
- Model validation
- Performance comparison
- Experimental evaluation
- Phase 2 report
- Final project report
- Final presentation

- Project demonstration video
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Final Deliverable of Entire Project

An AI-powered supply chain platform that can:

1. Understand supply chain questions.
2. Retrieve organizational knowledge.
3. Provide trusted answers with citations.
4. Analyze operational data.
5. Recommend decisions.
6. Explain recommendations.
7. Support supply chain managers in daily operations.

The project evolves from:

"What does the organization know?" (Phase 1)

to

"What should the organization do?" (Phase 2)

This progression makes the project strongly aligned with Industrial Engineering, Supply Chain Management, Operations Research, and AI-based Decision Support Systems.